

Is This Evidence Credible?

Module 2 · Day 4, Session 1: A Quality-Assurance Deep Dive

3ie

0 · Sign-off

From the analyst to the report

Yesterday the analyst admitted things out loud. This is the written report that goes to the minister.

Does it say what the analyst admitted?

Before you open it

1,000

AED per business per year · the authority's bar



Would you sign off on this report, as written, for national scale-up? Yes / No / Not without more answers. Vote now, before reading.

Four pages, confident and well written, with real strengths. Does it support its own claim?

1 • The report

What is in front of you

The report, section by section

Section

1. Executive summary
 2. The programme
 3. Evaluation design
 4. Data
 5. Results
 6. Limitations
 7. Conclusions and recommendations
- Annexes A and B

Read only the executive summary for now. Do not read ahead.

What the report claims

This evaluation finds that the programme reduced waste-management costs by **approximately 1,000 AED per business per year**, and that the estimate is statistically significant. The finding is robust across four independent estimation methods... The programme therefore meets the threshold for scale-up.

Four claims in three sentences. Note each one, because you will rate them separately:

- a **number**, described as approximately 1,000
- a claim of **statistical significance**
- a claim of **robustness** across four methods
- a claim that the **threshold is met**

Read the summary and rate it

On your rating sheet: **the executive summary.**

Give it a colour. Then write the one reason that decided it. If you rate it amber or red, write what would settle it.

Five minutes. Work alone. The point is your own reading, before anyone else's.

If you found nothing wrong, say so. Hold your colour and we will come back to it.

2 · The four areas, applied

The checklist, on paper this time

The four areas from yesterday

Area	What you are testing
Methodology	Is the design capable of answering the question at all?
Assumptions	What has to be true, and was it actually tested?
Data quality	Where did the numbers come from, and who is missing?
Conclusions	Does the claim follow, or does it outrun the evidence?

Yesterday the analyst answered these out loud. Today the report has to answer them. Your rating sheet names the area to check for each section.

Which of the four areas can a report make look strong without actually being strong?

The five questions, and what a report can hide

1. What is the comparison, and how was it chosen?
2. What has to be true for this to be causal, and did you test it?
3. Where did the numbers come from, and who is missing from them?
4. How big is the effect, and how sure are you of it?
5. Does the conclusion follow, or does it outrun the evidence?

None of the five asks whether the report sounds convincing.

What the rating means

| **Green.** Credible as it stands. You would rely on this section without further work.

| **Amber.** Credible with a specific caveat. Name the caveat and say what would settle it.

| **Red.** Not credible. The problem undermines a claim made in this section.

| What single fact would move your amber to green or red?

3 · Section by section

Your turn: rate the rest

20 minutes in pairs: rate Sections 3 to 7 from the report itself, on the rating sheet. Do not wait for the slides.

For each section: a colour, the one reason that decided it, and, if amber or red, what would settle it. The rating sheet names the area to check.

Evaluation design

The parallel trends assumption underlying the difference-in-differences design is the standard assumption in this literature and is satisfied by construction in programmes of this type, since treatment and control neighbourhoods were selected from the same administrative area and are subject to the same economic conditions.

Your colour for Section 3? What does parallel trends actually require?

That the two groups **would have moved in parallel without the programme**. Sharing an area makes a common shock likely. It does not show the groups were on the same path, and no pre-trend test is reported.

The groups are also the wrong ones: the DiD compares enrolled with non-enrolled businesses inside treatment neighbourhoods.

Data

Recycling compliance... is missing for 8,570 of the 19,826 observations. These gaps reflect the inspection schedule...

...the survey instrument separated expenditure on collection from expenditure on treatment... This change improves the precision of the measure...

Your colour for Section 4? Two disclosures: which is handled well?

Handled well: the instrument change is *disclosed*. Many reports would not mention it.

Not handled: the change is never tested for its effect on the outcome, and the report never says which businesses lack compliance data.

Results

All four estimates are negative and statistically significant... The largest estimate, from the randomised comparison, indicates a cost reduction of 1,014 AED per business per year. The randomised comparison is our headline result.

Your colour for Section 5? Four estimates are given. Why is this one the headline?

The report's reason: the randomised comparison covers all eligible businesses, the population that would be scaled up. That reason is defensible. It is also the largest estimate, and the only one above 1,000.

The table you should look at twice

Table 1 of the report, with two columns added

Method	Estimate (AED)	95% CI	Point estimate reaches 1,000?	Whole interval reaches 1,000?
Difference-in-differences	-816	[-879, -753]	No	No
Regression discontinuity	-905	[-992, -818]	No	No
Randomised comparison	-1,014	[-1,093, -935]	Yes	No
Matched comparison	-1,000	[-1,072, -927]	Yes, exactly	No

At the point estimate, two of the four reach 1,000. At the cautious end of the interval, none does. The preferred specification, named in Section 3, is 816.

Limitations

We note three limitations. First, the randomised comparison estimates the effect of being offered the programme, not of enrolling... Second, the regression discontinuity estimate is local to the eligibility cut-off... Third, self-reported expenditure may be subject to recall error...

Your colour for Section 6? Are these the right three?

All three are correct, and none of them threatens the conclusion.

The problems raised in Section 4 (the change in the survey instrument and the unexplained gap in the compliance data) are never mentioned again.

Conclusions

Recommendation 2. The subsidy rate should be maintained at its current level, since the evaluation provides no evidence that a lower rate would be less effective.

Recommendation 3. The eligibility cut-off should be reviewed, as the regression discontinuity results suggest that businesses just above the threshold may also benefit.

Your colour for Section 7? What was estimated about subsidy rates, and about businesses above the cut-off?

Nothing on either. Recommendation 2 turns an absence of evidence into a recommendation to do nothing. Recommendation 3 rests on a result about businesses *near* the cut-off, and even there the estimate was 905, below the threshold.

4 • The verdict

The same question

1,000

The bar, and the one estimate the summary chose to echo

Would you sign off on this report, as written? Vote again.



If your vote moved, name the section that moved it.

The most important problem

The results section. The headline is the one estimate that clears the threshold, and the report never says that its own preferred specification, named in Section 3, does not. A reader who stops after the executive summary would conclude the programme pays for itself. It does not, on the evidence reported here.

Which sections did you rate green, and why do they still hold?

The strengths are real

Before we close, the report's genuine merits:

- It **states the threshold up front**, so the reader knows the standard being applied.
- It **reports four methods** with their different populations named, rather than one number.
- It **discloses** the survey instrument change, which most reports would bury.
- It gives **confidence intervals for every estimate**, and an annex of robustness specifications.
- Its **limitations are technically correct**, even though they are the wrong ones.

Does any of that rescue the recommendation?

5 • Put AI to work

The prompt

Here is an evaluation report on a programme that subsidised waste-management technology for businesses. It found a cost reduction of approximately 1,000 AED per business per year and recommends national scale-up. Assess the credibility of this evidence and tell me whether the recommendation is sound.

You have just done this by hand, section by section. Run the prompt on the report yourselves, then compare the AI's verdict, and the example on the next slides, with yours. The report is on the course website: download Oct15_session1_report.docx from the landing page.

What came back

A response like this is possible, especially with a short or leading prompt:

Assessment: the evidence is credible and the recommendation is sound.

The report is unusually thorough for this kind of evaluation. Several points support its credibility:

Multiple methods. The use of four estimation approaches, all producing consistent results, is a strong indicator of robustness. When independent methods agree, the finding is unlikely to be an artefact of any single specification.

Large sample. With roughly 9,900 observations, the study is well powered and the estimates are precise.

What came back, continued

Transparency. The report discloses a change in the survey instrument between rounds and notes the gaps in the recycling compliance data. This level of disclosure suggests a careful and honest evaluation team.

Appropriate limitations. The limitations section correctly identifies that the randomised comparison estimates the offer effect and that the RDD estimate is local to the cut-off.

Recommendation. The headline estimate of 1,014 AED exceeds the 1,000 AED threshold, so **the programme meets the criterion for scale-up**. The recommendation is **supported by the evidence**.

Find the two claims the report does not support, then the larger problem. What did the AI you ran get right, and what did it miss?

What AI got, and what it could not get

What it did well

- Found the real disclosures and called them transparency
- Correctly described the offer-effect and locality limitations
- Reached a verdict, which is what was asked

What it missed

- It read “approximately 1,000” as if all four methods agreed. Only the matched comparison gives 1,000.
- It never noticed that the preferred specification, 816, fails the threshold.
- It treated the compliance gap as evidence of honesty.
- It did not ask why 1,014 is the headline.

Which of the four would slip past you in practice?

The same task, a better prompt

The same report claims a cost reduction of approximately 1,000 AED against a decision threshold of 1,000 AED. Do three things.

First, list every estimate the report gives, with its confidence interval, and say for each one whether it clears the threshold.

Second, identify any claim in the executive summary that is not supported by the results section, and quote both.

Third, tell me what the report does not tell me that I would need in order to decide.

What do the three instructions have in common?

Each asks for a comparison: estimates against the threshold, the summary against the results, the report against what a decision needs.

Two checks on any AI answer

- 1.** Ask the AI to ask you questions before it answers. It then has to find out what it does not know, such as which estimate is the preferred specification.
- 2.** Paste its answer into a fresh chat and ask what is wrong with it. A new chat has not committed to the first verdict.

6 · Take this back to your desk

Three habits for reading a report

- 1.** Tabulate the estimates yourself, with the threshold in its own column. Do not let the report choose which number you see first.
- 2.** Read the limitations section, then check whether it names the problems in the data section. If it does not, ask why they were dropped.
- 3.** Find the preferred specification and check whether the headline is it. When they differ, ask why.

Which of the three would have caught the most important problem in this report?

Next session

This week you have practised reading other people's evidence critically. In the next session you write your own: a policy brief, from findings to recommendation.

Bring the ratings you gave today. You will apply the same judgements to your own draft.